INCH-POUND

MIL-DTL-87255A(USAF)
6 June 2008

SUPERSEDING
MIL-V-87255 (USAF)
31 December 1990

DETAIL SPECIFICATION

VALVES, PRESSURE, ANTI-G SUIT, HIGH FLOW

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

MIL-DTL-87255 (USAF) is inactive for new design and is no longer used, except for replacement purposes.

1. SCOPE

1.1 Scope.

This specification covers a high-flow, pressure regulating valve for inflating anti-G suits and providing a sense line signal for a g-compensated oxygen regulator (if applicable). It also includes details of inlet and outlet fittings required to interface the valve to specific aircraft.

1.2 Classification.

The designations for the three valves are as follows:

a. VAU-45(V)1/A

b. VAU-45(V)2/A

c. VAU-45(V)3/A

Comments, suggestions, or questions on this document should be addressed to ASC/ENRS, 2530 Loop Road West, Wright-Patterson AFB OH 45433-7101 or emailed to Engineering.Standards@wpafb.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1660

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2. APPLICABLE DOCUMENTS

2.1 General

The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For (Inactive for New Design)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-1916	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.100	Engineering Drawing and Related Documentation Practices

(Copies of this document are available online at http://www.asme.org or from ASME, P.O. Box 2300, Fairfield, NJ 07007-2300.)

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Copies of this document are available online at www.asq.org or from the American Society for Quality, 600 North Plankinton Ave., Milwaukee WI 53203-2914.)

SAE INTERNATIONAL

SAE AS 4395 Fitting End, Flared, Tube Connection, Design Standard (DoD Adopted)

SAE AS 5132 Fitting End-Hose Connection, Design Standard (DoD Adopted)

(Copies of these documents are available online at http://www.sae.org or from SAE International, 400 Commonwealth Dr., Warrendale, PA 15096-0001.)

2.4 Order of precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article.

When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

Materials shall conform to applicable specifications and standards as specified herein and on applicable drawings. Materials which are not covered by specifications and standards, or specifically described herein, shall be reported to the acquisition activity and shall require contracting officer approval prior to use. Materials shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended.

3.2.1 Recycled, recovered, or environmentally preferable materials.

Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Metal parts.

All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

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3.2.2.1 Dissimilar metals.

Unless protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.2.3 Nonmetallic materials.

Any nonmetallic material that is adversely affected by continued use with air shall not be used.

3.2.3.1 Elastomer components.

Elastomer components, except silicone, shall be not more than 12 months old from the date of manufacture to the date of delivery to any government service or to any airframe or accessory manufacturer.

3.2.4 Protective treatment.

Materials shall be protected from deterioration in a manner that shall in no way prevent compliance with the performance requirements of this specification. The valve shall resist corrosion and deterioration due to fuels, salt fog, microorganisms, moisture, hydraulic fluid, lubricating oils, grease, ozone, or atmospheric conditions likely to be encountered in storage or normal service. Protective coatings which might crack, chip, or scale during normal service or storage life or under extremes of environmental conditions shall not be used.

3.3 Standard parts.

Military standard parts shall be used when they suit the purpose. Use of non-standard parts requires contracting officer approval prior to use.

3.4 Design and construction.

The valve shall be designed to pass air at a regulated pressure to the aircrew member's anti-G suit and to the sense line of the regulator (if applicable) and shall contain provisions for suit exhaust. The valve shall include provision for inlet and outlet fittings that incorporate swivel features. The fittings shall have a diameter and finish compatible with the aircraft inlet hose, anti-g suit delivery hose, and the outlet pressure sense line (if applicable). The outlet fitting shall be in accordance with figures 1, 2 or 3, as required. The outlet pressure shall be a function of the acceleration force applied to the valve in the $+G_z$ (see 6.4.1) direction. The design shall enable manual operation (press-to-test) of the valve and shall provide means to simulate acceleration forces (see 3.8.2 and 4.4.2). The valve shall be designed to perform as specified in 3.5. Moisture in the air source to the valve shall not adversely affect operation of the valve.

3.4.1 Installation.

The valve shall be designed such that it can be properly installed in the applicable aircraft. This includes proper alignment with the mounting brackets and connections with the inlet and outlet hoses. The existing control panels shall also be accommodated. There shall be no interference with aircraft structure. Minimum requirements for installation are given on figures 1, 2, and 3. Proper installation shall be demonstrated as specified in 4.5.1.1.1. Should there be a conflict between the installation demonstration specified in this document and figures 1, 2, and 3, the installation demonstration shall take precedence.

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3.4.2 Relief valve.

The valve shall include a pressure relief valve that shall not allow the valve outlet pressure to the anti-g suit and the sense line (if applicable) to exceed 13 psig.

3.4.3 Lubrication.

The valve shall be designed and constructed in such a manner that no maintenance lubrication shall be required.

3.4.4 Inlet screen.

The valve shall contain a corrosion-resistant steel, 30 by 30 maximum mesh, 0.010 inch minimum wire diameter inlet screen. All inlet air shall pass through the screen.

3.5 Performance.

All specified performance requirements shall be based on the use of air.

3.5.1 Minimum operating acceleration force.

The minimum equivalent acceleration force required to open the valve to permit flow of 0.01 liter per minute shall be between 1.25 and 1.50 G_z (see 4.5.2).

3.5.2 Outlet pressure regulation.

For a constant applied acceleration force, the outlet pressure schedule during testing (see 4.5.3) shall be within the range specified in table I and displayed graphically on figure 4 for both gradual onset/offset (<= 0.1 G_z/sec) and rapid onset/offset (= 4.0 G_z/sec). Outlet pressure shall be met within 2.0 seconds of reaching the constant force when accelerating and within 5.0 seconds of reaching the constant force when decelerating. The outlet pressure shall not fluctuate more than ±0.1 psi at any given constant acceleration force.

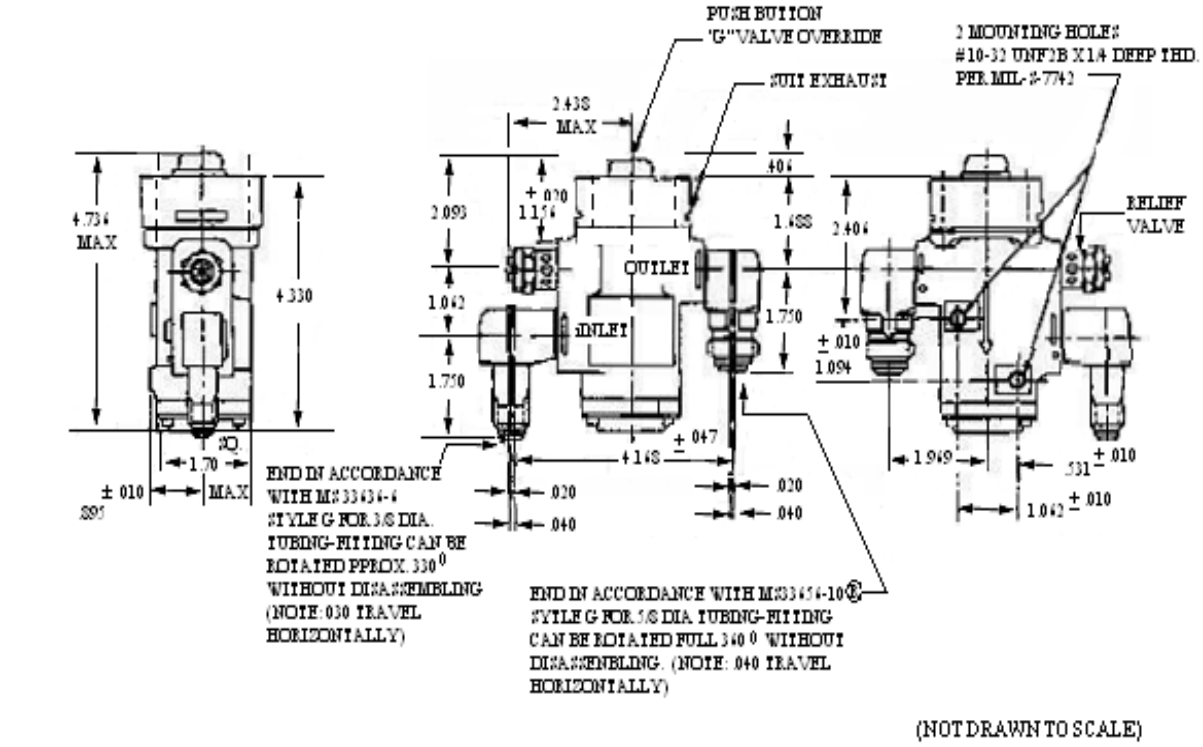
TABLE I. Outlet pressure.

G-Range ($G_{z)}$	Maximum Pressure (psig)	Minimum Pressure (psig)
0 – 1.80	-	0
1.80 – 2.00	-	G_z – 1.80
2.00 – 4.00	-	1.40 +G_z – 2.60
4.00 – 9.33	-	1.50 +G_z – 3.00
> 9.33	-	11.00
1.25 – 4.00	1.45 +G_z – 1.80	-
4.00 – 10.00	1.50 +G_z – 2.00	-
>10.00	13.00	-

3.5.3 Leakage.

The leakage to the ambient shall not exceed 1.0 liters per minute. There shall be no leakage to the outlet (see 4.5.4).

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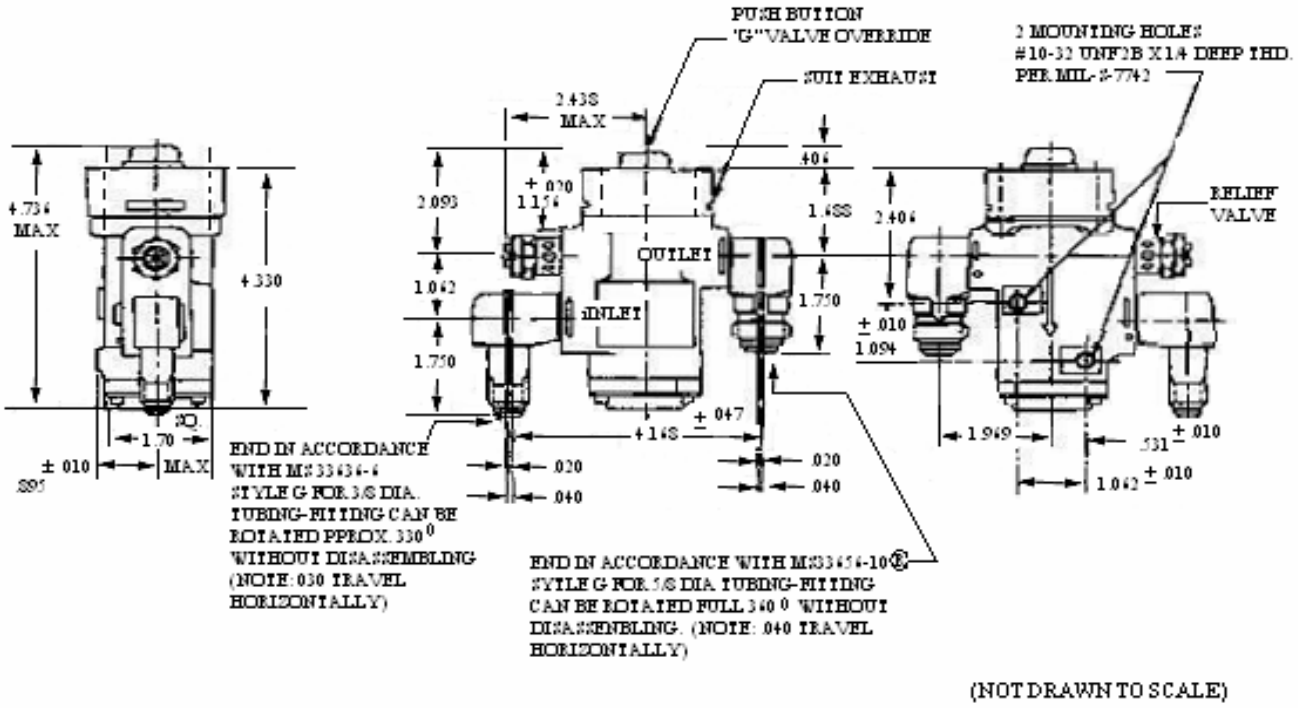


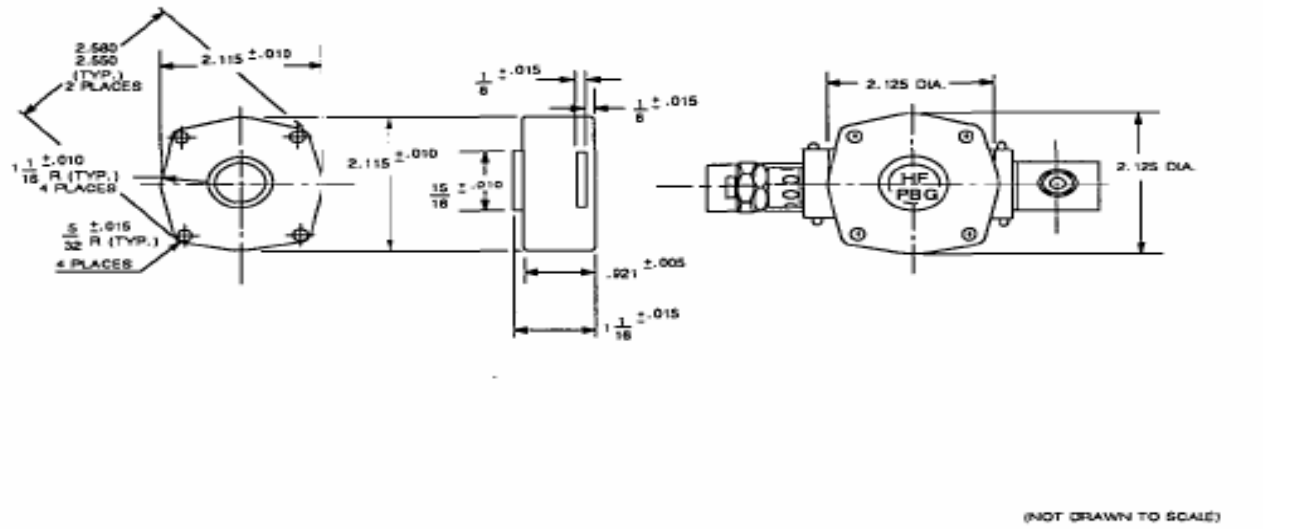
NOTES:

1. Dimensions are in inches.
2. MS33656 has been cancelled. Future acquisitions for this material should refer to SAE AS4395.
3. MIL-S-7742 is Inactive for New Design.

FIGURE 1. Aircraft interface dimensions for VAU-45(V)1/A.

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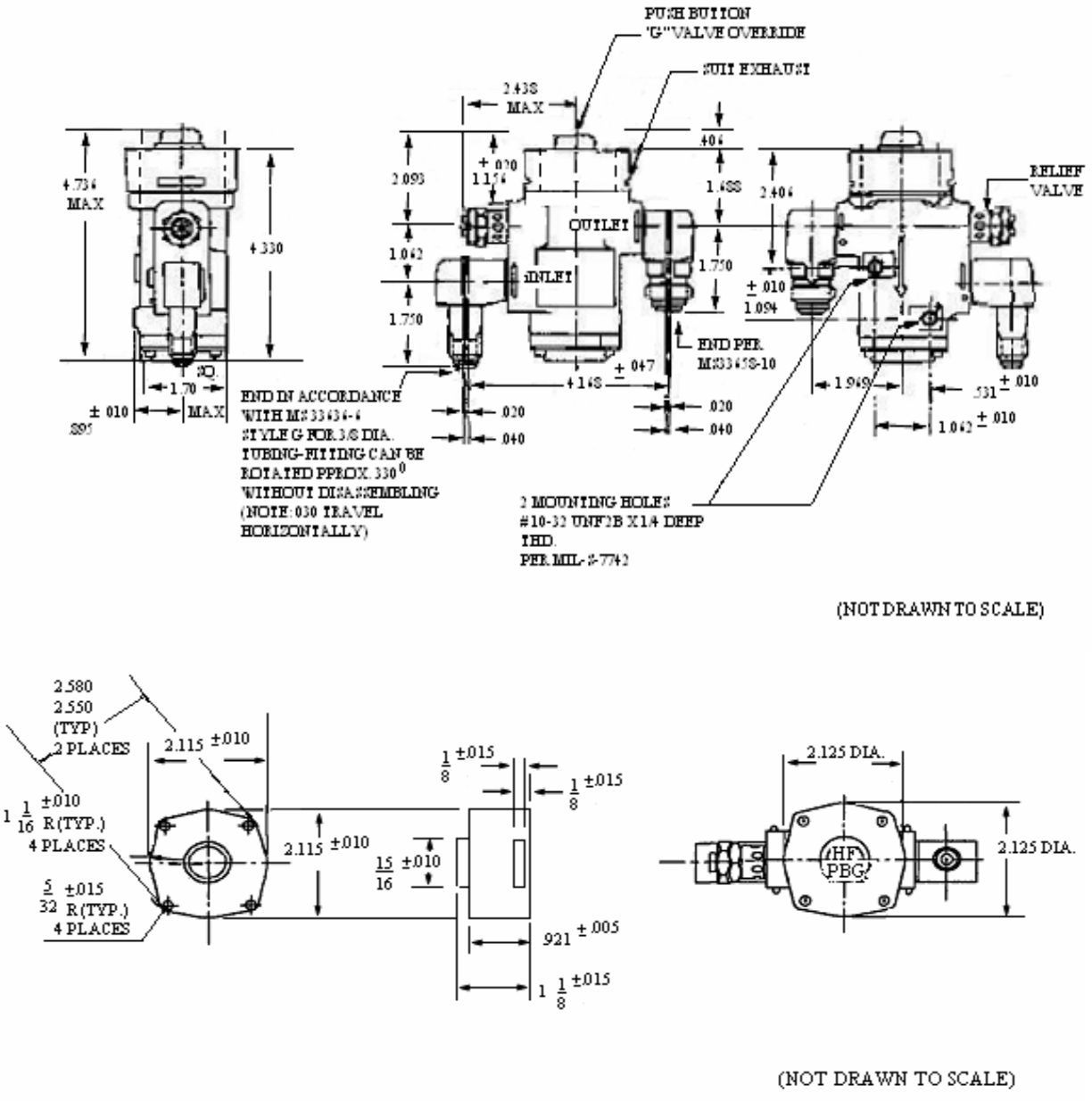


NOTES:

1. Dimensions are in inches.
2. MS33656 has been cancelled. Future acquisitions for this material should refer to SAE AS4395.
3. MIL-S-7742 is Inactive for New Design.

FIGURE 2. Aircraft interface dimensions for VAU-45(V)2/A.

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NOTES:

1. Dimensions are in inches.
2. MS33656 and MS33658 have been cancelled. Future acquisitions for this material should refer to SAE AS4395 and SAE AS5132, respectively.
3. MIL-S-7742 is Inactive for New Design.

FIGURE 3. Aircraft interface dimensions for VAU-45(V)-3/A.

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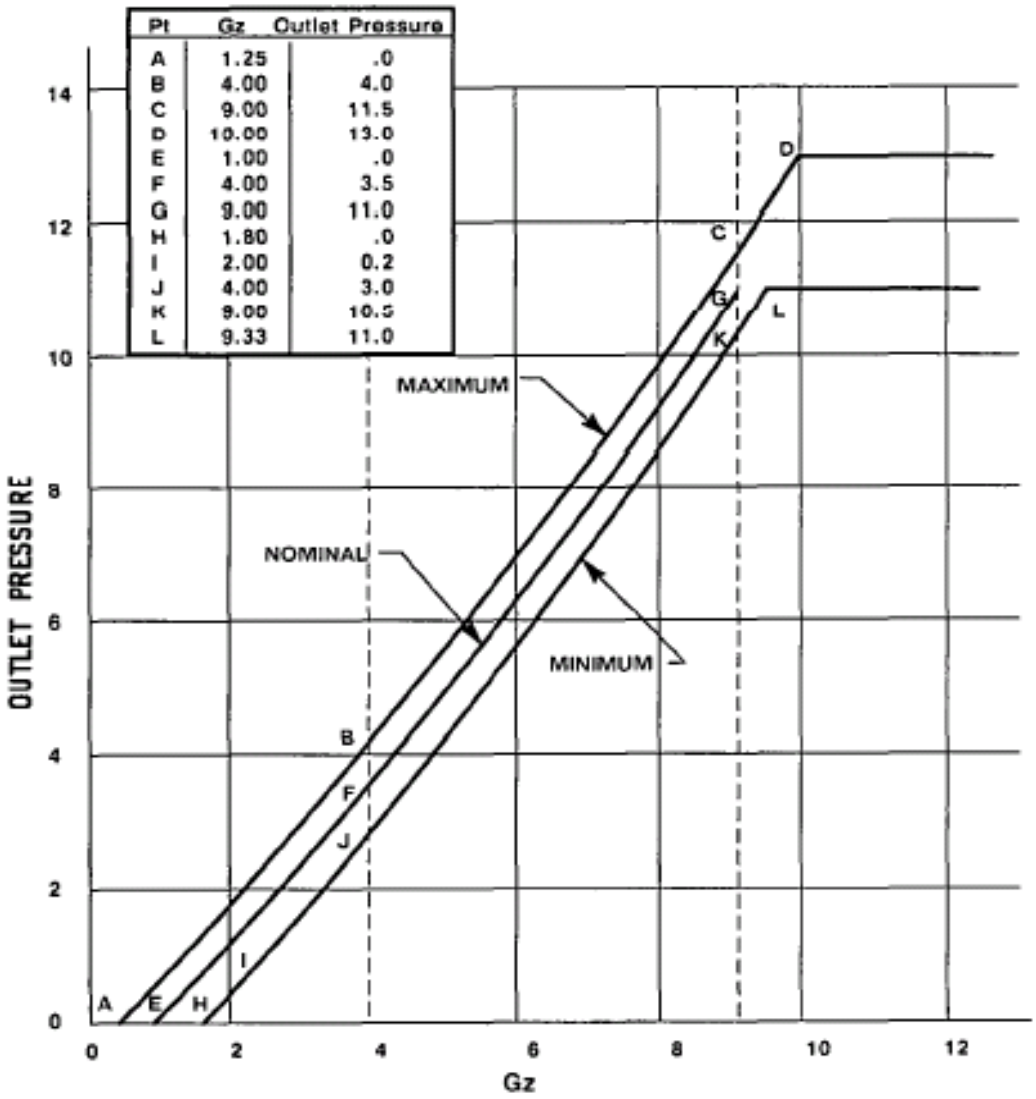


FIGURE 4. Anti-g valve outlet pressure vs G_z.

3.5.4 Anti-chatter.

The valve shall not chatter (see 4.5.5 and 6.4.2).

3.5.5 Outlet pressure response

3.5.5.1 Response time.

When the acceleration force is being applied by a centrifuge or an aircraft, the outlet pressure shall be in accordance with table I (see figure 4) within 2 seconds after the application of the force. When the acceleration force is being decreased, the pressure shall be in accordance with table I (see figure 4) within 5 seconds after decreasing the force. Additionally, the outlet pressure shall change in a smooth continuous manner without oscillations or step responses.

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3.5.5.2 Test limit response time.

During testing using a step response input to simulate the G force the tank pressure shall have the response time given by table II (see 4.5.6).

3.5.6 High temperature exposure.

The valve shall operate satisfactorily after completion of the high temperature test at 250 ±5 °F (see 4.5.7).

3.5.7 Low temperature exposure.

The valve shall operate satisfactorily after completion of the low temperature test at -65 ±5 °F (see 4.5.8).

3.5.8 Vibration.

The valve shall operate satisfactorily and shall show no evidence of mechanical or material failure after completion of the vibration tests (see 4.5.9).

3.5.9 Salt fog.

The valve shall show no evidence of corrosion of mechanical failure after completion of the salt fog test (see 4.5.10).

3.5.10 Internal corrosion.

The valve shall show no evidence of corrosion of mechanical failure after completion of the salt fog test (see 4.5.11).

3.5.11 Centrifuge testing.

This testing shall be done using valves of the same design as those that passed all other performance tests in 3.5. Centrifuge testing shall include an anti-chatter test and performance testing (see 4.5.12).

3.5.12 Endurance.

The valve shall have a mean-cycle-between-failure of 15,030 cycles (see 6.4.3) and shall pass the endurance test (see 4.5.13).

3.5.13 Disassembly and examination.

The valve shall not show excessive evidence of deterioration or wear due to testing which might affect the valve performance (see 4.5.14).

3.6 Interchangeability.

All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable with each other with respect to installation and performance. The item identification and part number requirements of ASME Y14.100 shall apply.

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TABLE II. Test response time.

Force Applied From	Tank Pressure	Response Time (Max)
1 to 10 G_z	As specified on figure 4 for 10 G_z	1.6 sec
10 to 1 G_z	0	2.7 sec
1 to 6 G_z	As specified on figure 4 for 6 G_z	1.6 sec
6 to 1 G_z	0	2.2 sec

3.7 Weight.

The total weight of the valve shall not exceed 3 pounds.

3.8 Identification of product.

The valve shall be marked for identification in accordance with MIL-STD-130 and shall include at least the following information:

Valve, Pressure, Anti-G Suit, 1/
Manufacturer's Part No.
Manufacturer's Identification
U.S. MXXXXX-X (see 6.5)
Date of Manufacture: MFD (Month)(Year) 2/

1/ Enter the applicable designation (VAU-45(V)1/A, VAU-45(V)2/A, or VAU-45(V)3/A here.

2/ The manufactured date may be stamped on the valve rather than marked on the nameplate and shall be as permanent as the normal life expectancy of the valve.

3.8.1 Ports.

All inlet and outlet ports shall be clearly identified by permanent markings.

3.8.2 Press-to-test button.

The valve shall have a permanent marking on the top of the press-to-test button to identify it as a "HI FLOW" for the VAU-45(V)1/A and as "HF PBG" for VAU-45(V)2/A and VAU-45(V)3/A. The press-to-test button shall enable manual simulation of acceleration forces.

3.8.3 Direction of acceleration force.

The valve shall be marked with an arrow in accordance with figures 1, 2, or 3 denoting the direction the influencing acceleration force shall be applied.

3.8.4 Workmanship.

The valves shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

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4. VERIFICATION

4.1 Classification of inspections.

The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

First article inspection shall be as specified by the contracting activity. Table III consists of all the examinations and tests of this specification.

4.2.1 First article samples.

The samples shall be of identical construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these valves, from contract to contract, first article inspection on the new contract may be waived at the discretion of the contracting activity (see 6.2). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the conformance inspection.

4.2.1.1 Disposition of sample units.

Upon completion of the first article inspection, all the applicable inspection reports, sample valves, and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One of the approved first article inspection samples of the valves will be returned to the contractor for use in monitoring production. The other valves will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract or purchase order.

4.3 Conformance inspection.

Conformance inspection shall consist of the examination and tests specified in table IV. The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-1916 or ANSI/ASQ Z1.4. The lot from which the valve samples were selected shall be rejected if any sample fails to comply with any requirement specified. A rejected lot shall not be resubmitted except with the approval of the contracting officer.

4.3.1 Endurance acceptance criteria.

The endurance requirements of 3.5.12 shall be verified by lot sampling tests to be conducted at the contractor's facility in accordance with the criteria specified in 4.5.13. The acceptance criteria for the first sampling test is 30,060 cycles with zero failures. The 30,060 cycles shall be accumulated on two units, 15,030 cycles on each unit, selected from the first 10 units produced under this specification. One unit out of every subsequent production lot of 500 shall be subjected to 15,030 cycles to the criteria stated above with zero failures as the acceptance criteria. Corrective actions for all failures shall be determined and recommended to the procuring activity. In the event of non-compliance (reject decision), or when a pattern failure occurs, the contractor shall immediately notify the procuring activity. Changes of specified performance or required endurance characteristics of the valves are not corrective action.

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4.3.2 Sampling.

4.3.2.1 Inspection lot.

4.3.2.1.1 Valve.

An inspection lot size shall be expressed in units of valves, manufactured essentially under the same conditions and from the same materials and components. The sample unit shall be one valve.

4.3.2.2 Sampling for tests and inspections of valves.

The sample size, acceptance criteria, tests, and inspections required for the valves shall be as specified in table II.

4.4 Test conditions

4.4.1 Environmental conditions.

Unless otherwise specified, all tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions shall be 29.92 inches of mercury and 70 °F.

4.4.2 Acceleration force.

When conducting the tests specified herein, the acceleration force (see 6.4.1) shall be applied to the valve in the direction specified on figures 1, 2, or 3. Weights or any other means may be used to simulate the acceleration force provided the weight of the sensing mass of the valve shall be controlled, and verified by 100 percent inspection. In this case the total G force shall be 1 G_z plus the amount simulated by the test weights. The acquisition activity reserves the option to verify the performance of the valve on a centrifuge as specified in 4.5.12.

4.4.3 Test medium.

Unless otherwise specified, the test medium used in testing the valves shall be air or nitrogen. The moisture content of the air shall not exceed 0.02 milligram per liter. The nitrogen shall conform to type I, grade B, class 1, of A-A-59503. When nitrogen is used as a test medium, appropriate density corrections shall be made in determining that the test requirements are met.

4.4.4 Test equipment

4.4.4.1 Tank.

When specified, a rigid tank having a volume of 10 ±0.05 liters shall be used. The tank shall be connected to the outlet port of the valve as specified in 4.4.4.2 and shall provide a connection for the measurement of pressure in the tank. Unless otherwise specified, the sensing line port (if applicable) shall be capped for all tests.

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4.4.4.2 Test set-up.

The tank specified in 4.4.4.1 shall be connected to the valve by piping which simulates the resistance characteristics of the anti-G suit hose, pressure sense line (if applicable), and applicable aircraft quick disconnect fittings. As a minimum, inlet and outlet pressure and the applied G force (if applicable) shall be measured. Figure 5 can be used as a guide. The pressure sensors shall have a flat frequency response from 0.0 to 1000 Hz.

TABLE III. First article inspection.

Inspection	Requirement	Examination or Test Method
Visual examination		4.5.1
Minimum operating acceleration force	3.5.1	4.5.2
Outlet pressure regulation	3.5.2	4.5.3
Leakage	3.5.3	4.5.4
Anti-chatter	3.5.4	4.5.5
Response time	3.5.5.1	4.5.6
High temperature exposure	3.5.6	4.5.7
Low temperature exposure	3.5.7	4.5.8
Vibration	3.5.8	4.5.9
Salt fog 1/	3.5.9	4.5.10
Internal corrosion 1/	3.5.10	4.5.11
Centrifuge testing	3.5.11	4.5.12
Endurance 1/	3.5.12	4.5.13
Disassembly and examination	3.5.13	4.5.14

1/ This test shall be considered destructive. Valves subjected to this test shall not be considered or shipped as part of the contract or purchase order. This test shall be conducted on new valves ready for delivery to the Government.

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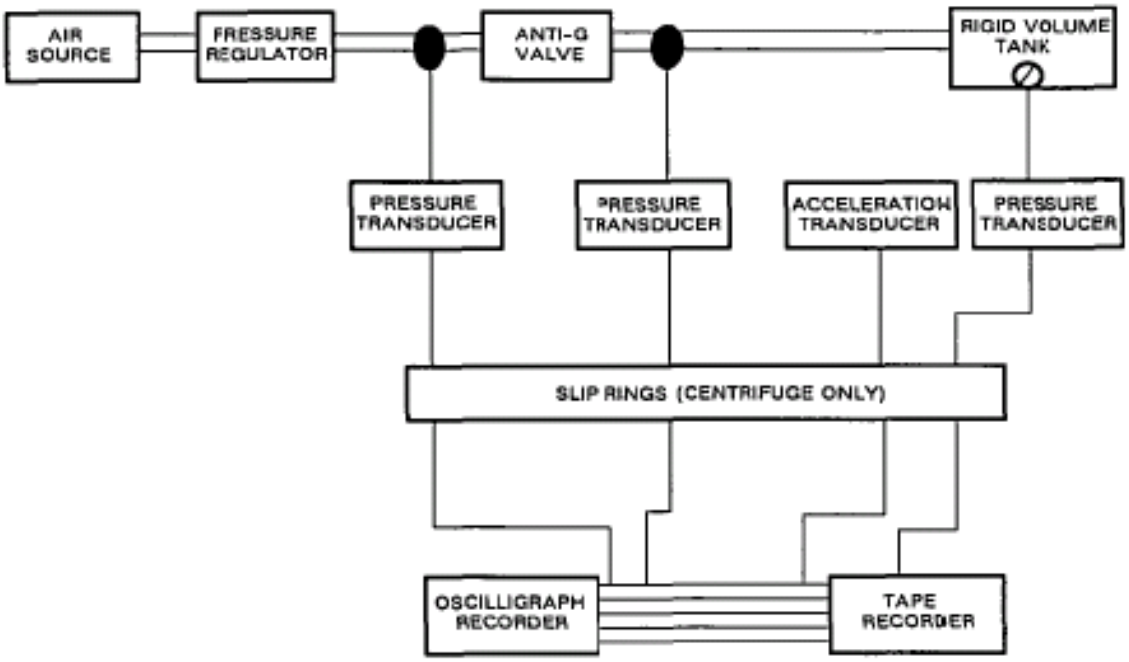


FIGURE 5. Test set-up.

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TABLE IV. Conformance inspection.

Test or Inspection	Test Method	Sample Size	Classification of Characteristics	Acceptance Criteria
Visual examination. Inspect for relief valve safety wired	4.5.1	Every valve for critical defects	Critical	Reject all valves with any critical defects.
		Inspection level II for minor defects	Minor	An acceptable level of 2.5 defects per hundred units for minor defects..
Dimensions and configuration	4.5.1.1.1	Inspection level S-2	Major	Acceptance number of 0,, rejection number of 1.
Minimum operating acceleration force	4.5.2	Every valve	Major	Acceptance number of 0, rejection number of 1.
Outlet pressure regulation	4.5.3	Every valve	Critical	Acceptance number of 0, rejection number of 1.
Leakage	4.5.4	Every valve	Major	Reject all defective units.
Anti-chatter	4.5.5	Inspection level I	Major	Acceptance number of 0, rejection number of 1.
Response time	4.5.6	Every valve	Critical	Reject all defective units.
Endurance 1/	4.5.13	See 4.3.1	Major	See 4.3.1
1/ This test shall be considered destructive. Valves subjected to this test shall not be considered or shipped as part of the contract or purchase order. This test shall be conducted on new valves ready for delivery to the Government.				

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4.5 Inspection methods

4.5.1 Visual examination

4.5.1.1 Valve.

Each valve selected as a sample unit from the lot shall be thoroughly examined visually and checked dimensionally for defects to determine conformance to this specification with respect to materials, design, construction, marking, and workmanship. Defects shall be classified in accordance with table V.

4.5.1.1.1 Dimensions and configuration.

The valves shall be examined for defects in dimensions and configuration. Any dimensions or configuration not conforming to the requirements of the applicable assembly number of figures 1, 2 or 3, as applicable, shall be classified as a major defect.

4.5.2 Minimum operating acceleration force.

At the start of this test the press-to-test button shall be depressed to ensure air flow at the outlet. No air flow at the outlet with the button fully depressed shall be cause for failure of this test. The valve shall be subjected to inlet pressures of 30, 70, 150, and 300 psig. The acceleration force required to open the valve at each inlet pressure shall be as specified in 3.5.1.

TABLE V. Classification of defects for visual examination of the valve.

Critical	Minor
1. Material imperfections – foreign matter embedded.	201. Marking – missing, insufficient, incorrect, permanent
2. Surface – unclean, rough, misaligned, or containing crack, nicks, or other flaws.	
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning pan that works with difficulty.	
7. Faulty workmanship or other irregularities.	
8. Linkage interference or component binding.	

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4.5.3 Outlet pressure regulation.

A tank shall be connected as specified in 4.4.4 during this test. A source of pressurized gas shall be connected to the valve inlet port. The inlet gas pressure shall be increased and maintained at 30 psig while equivalent acceleration forces of 2, 4, 6, 8, 9, 10, 9, 8, 6, 4, 2, and 1 G_z shall be applied in this order to the valve. Each force shall be maintained for at least 10 seconds before the next force shall be applied. The inlet pressure shall be increased to 70, 150 and 300 psig, and the test repeated at each of these inlet pressures. The outlet pressure shall be as specified in 3.5.2.

4.5.4 Leakage.

With a force of 1 G_z applied to the valve, the inlet flow to the valve at an inlet pressure of 30, 70, 150, and 300 psig shall not exceed that specified in 3.5.3

4.5.5 Anti-chatter.

A tank shall be connected as specified in 4.4.4 during this test. A source of pressurized gas shall be connected to the valve inlet connection. The inlet pressure shall be increased and maintained at 30, 70, 150 and 300 psig. At each pressure level the valve shall be fully opened manually, by pressing the press-to-test button, and held open to determine that there is no evidence of chatter when there is air flow through the relief valve. Additional anti-chatter testing shall be as detailed in 4.5.12.2.5.

4.5.6 Response time.

A tank shall be connected as specified in 4.4.4 during this test. A source of pressurized gas shall be connected to the valve inlet connection and the pressure increased to and maintained at 70 psig. A pressure transducer shall be connected to the tank to allow tank pressure to be recorded as a function of time. Acceleration forces of 6, 1, 10 and 1 G_z shall be applied in this order as a step input to the valve. Each force shall be maintained for at least 10 seconds before the next force shall be applied. Tank pressure shall be recorded as a function of time for at least 5 seconds following the application of each force level. The time interval from the time of application of each force level to the time at which the tank pressure reaches the value specified in 3.5.5.2 shall not exceed the response time specified in 3.5.5.2.

4.5.7 High temperature exposure.

The pressure regulating valve shall be conditioned at a temperature of 250 ±5 °F for eight hours. After this period, the valve shall be immediately subjected to and pass the tests specified in 4.5.2, 4.5.3, 4.5.4, 4.5.5, and 4.5.6; except that the minimum operating force can be between 1.25 and 2.0 G_z.

4.5.8 Low temperature exposure.

The pressure regulating valve shall be stabilized at a temperature of -65 ±5 °F. After this period, the valve shall be warmed to -40 ±5 °F and immediately subjected to and pass the tests specified in 4.5.2, 4.5.3, 4.5.4, 4.5.5 and 4.5.6; except that the minimum operating force can be between 1.25 and 2.0 G_z.

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4.5.9 Vibration

4.5.9.1 Random vibration.

With the valve connected to the test equipment described in 4.4.4, with an inlet pressure of 300 psig, the valve shall be exposed for one hour at each simulated G level to the random vibration described below in each of three mutually orthogonal axes. After this test, there shall be no damage to the valve and the valve performance shall be verified as described by 4.5.2 through 4.5.6, except that the minimum operating force can be between 1.25 and 2.0 G_z. The random vibration envelope (see figure 6) shall be as follows:

0.04 g^2/Hz from 15 – 100 Hz

-4 db/octave slope from 100 – 140 Hz

0.025 g^2/Hz from 140 – 540 Hz

0.072 g^2/Hz from 540 – 900 Hz

-6 db/octave from 900 – 2000 Hz

4.5.9.2 Resonance.

With the test equipment and anti-g valve set up as described in 4.5.9.1, and using a sinusoidal vibration at 0.02 inches or ±2 G_z double amplitude whichever shall be less; a resonance search shall be conducted between 5 and 2000 Hz in three mutually orthogonal axes to identify resonant frequencies. The valve shall then be subjected to any and all resonant frequencies for seven and one half minutes each. There shall be no damage to the valve and the valve shall meet the requirements of 3.5.1 through 3.5.5 before and after this exposure. After this test, there shall be no damage to the valve and the valve performance shall be verified as described by 4.5.2 through 4.5.6, except that the minimum operating force can be between 1.25 and 2.0 G_z.

4.5.9.3 Gunfire vibration.

With the test equipment and anti-g valve operating as described in 4.5.9.1, the valve shall be subjected to the following vibrations along each of three mutually orthogonal axes. After this test, there shall be no damage to the valve and the valve performance shall be verified as described by 4.5.2 through 4.5.6, except that the minimum operating force can be between 1.25 and 2.0 G_z.

4.5.9.3.1 Excitation.

The excitation shall consist of four phase uncorrelated sinusoids superimposed over a random vibration background. The sinusoids shall be at frequencies representing the basic gunfire rate orders; 100 Hz (f_1), 200 Hz (f_2), 300 Hz (f_3), and 400 Hz (f_4). Synchronously, the frequencies shall be swept logarithmically over a bandwidth equal to ±0.2 times f1 f2, f3, and f4, beginning at the low frequency and ending at the high frequency limit. The sweep time for the combination sine sweep and random vibration shall be 60 minutes per axis. The instantaneous peaks of the random vibration may be limited to 2.5 times the root mean square spectral acceleration level. Spectrum frequencies, frequency-function levels, and location-function levels for the combined random and sinusoidal vibration tests shall be as contained on figure 7. After this test, there shall be no damage to the valve and the valve performance shall be verified as described in 4.5.2 through 4.5.6, except that the minimum operating force can be between 1.25 and 2.0 G_z.

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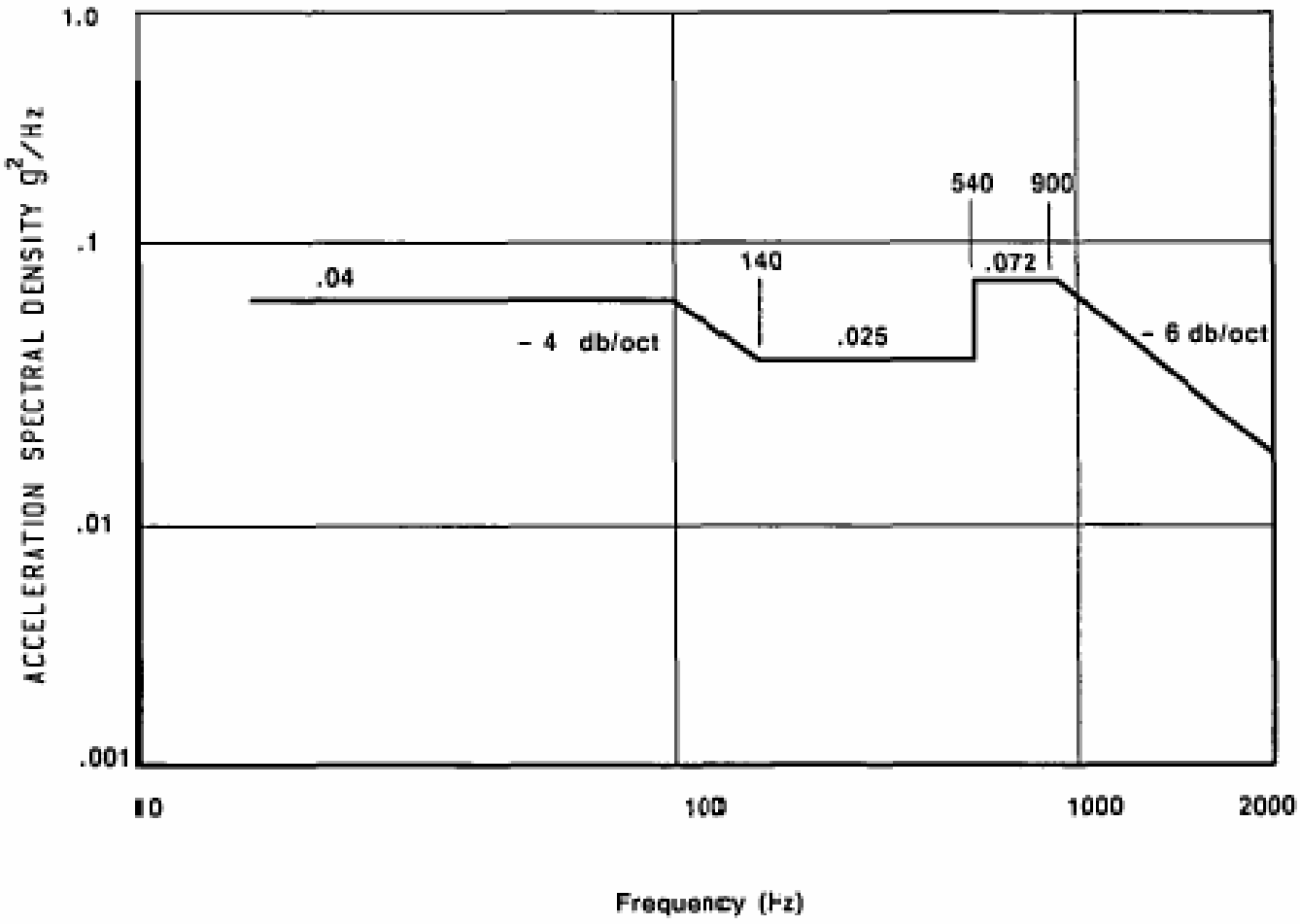


FIGURE 6. Random vibration versus frequency.

4.5.9.3.2 Resonance dwell.

The anti-g valve shall be subjected to eight resonance dwells per orthogonal axis, each lasting five minutes. The excitation level at the dwell frequencies shall be as described below and shown on figure 7. Background random excitation shall not be required for the following gunfire resonance tests. Upon completion of these tests, the valve performance shall be verified as described in 4.5.2 through 4.5.6, except that the minimum operating force can be between 1.25 and 2.0 G_z.

4.5.9.3.2.1 Frequency band.

One dwell shall be done in each of the frequency bands corresponding to ±10 percent of each of the above gunfiring frequencies (f_1, f_2, f_3, and f4). If no resonance is found in a gunfire frequency band, the dwell shall be performed at the center frequency of the band.

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4.5.9.3.2.2 Dwells.

Dwells shall be conducted in the frequency range from 100 to 1000 Hz at the four most significant resonant frequencies from 4.5.9.2.

4.5.10 Salt fog.

The valve shall be subjected to a salt fog test in accordance with method 509.4 of MIL-STD-810. The valve shall be examined for evidence of corrosion to all exposed surfaces. If the valve exhibits any corrosion indicative of dissimilar metals or corrosion that could preclude normal operation or create any unsafe condition, the valve shall be rejected. The valves shall be subjected to and pass the tests described in 4.5.2 through 4.5.6; except that the minimum operating force can be between 1.25 and 2.0 G_z and the outlet pressure can vary by an additional ±1.0 psi.

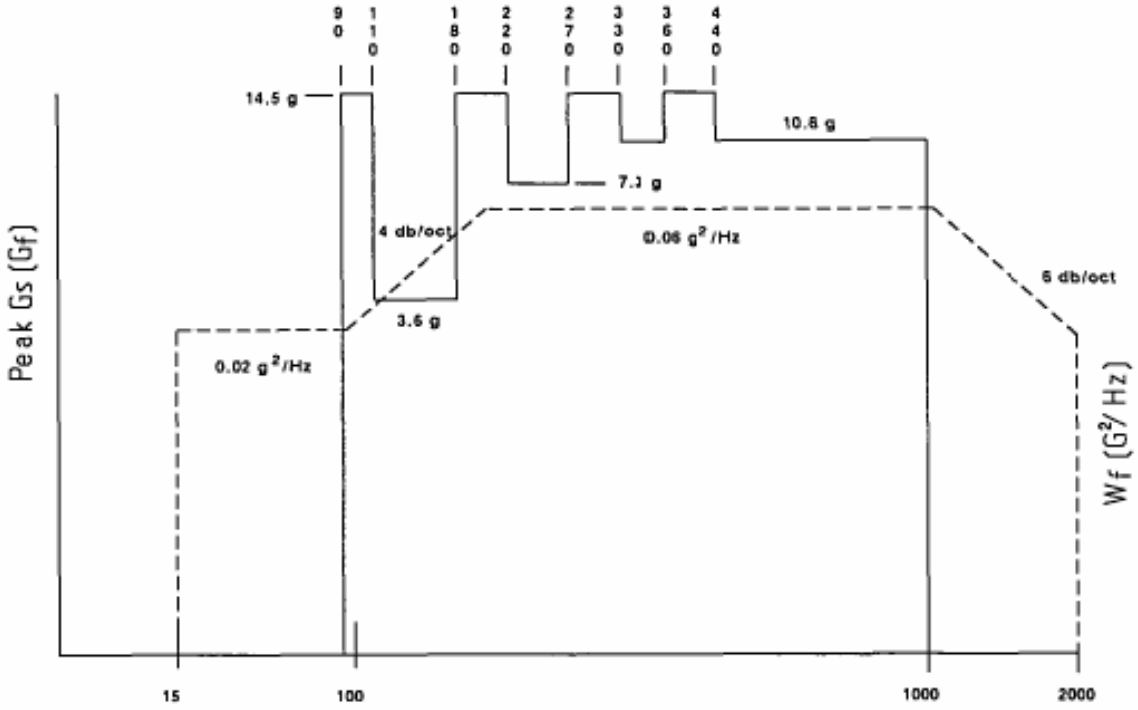


FIGURE 7. Combined random and sinusoidal vibration for gunfire environment.

4.5.11 Internal corrosion.

With the valve maintained at a temperature of 135 ±5 °F, a salt fog in accordance with the salinity requirements of MIL-STD-810, method 509.4, at a temperature of 135 ±5 °F, shall be passed through the valve in the normal direction of flow at approximately three flow cycles per minute for a duration of at least 15 minutes. The rate of flow of salt fog shall be no less than 30 liters per minute. This procedure shall be performed at least 10 times with the valve dried after each 15 minute period. After completion of the above tests, the valve shall be subjected to and pass the tests specified in 4.5.2 through 4.5.6; except that the minimum operating force can be between 1.25 and 2.0 G_z and the outlet pressure can vary by an additional ±1.0 psi.

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4.5.12 Centrifuge testing.

Centrifuge testing shall be conducted at a facility approved by the Government. The following shall be capable of being recorded.

a. G_z: 1 to 10 G with an accuracy of ±0.1 G

b. Tank or anti-g suit pressure: 0 to 15 psig, with an accuracy of ±0.1 psig

c. Valve outlet pressure: 0 to 15 psig, with an accuracy of ±0.1 psi

d. Source pressure: 0 to 300 psig, with an accuracy of ±10 psig

4.5.12.1 Sampling rates and calibrations.

The sampling rates and calibrations shall ensure compliance with the performance requirements of this specification. Both pre-test and post-test calibrations shall be required. The valve shall be mounted in radial line with the accelerometer, and connected in accordance with 4.4.4.2 to a tank as specified in 4.4.4.1. The test medium shall be as specified in 4.4.3 and the test set-up shall be in accordance with 4.4.4.2, with the exception of 4.5.12.2.5 which includes instructions for set-up.

4.5.12.2 Tests.

The following tests shall be performed:

4.5.12.2.1 Test profile 1.

With the valve inlet pressure maintained at 30 psig, the centrifuge shall be accelerated from 1.2 G_z to 10 G_z, held at 10 G_z for at least 10 seconds, and then decelerated back to 1.2 G_z. The acceleration and deceleration (onset/offset) rates shall be 0.1 G_z per sec. This shall be as shown on figure 8. G_z level at pressure initiation, the valve outlet pressure versus G_z level, and the relief valve opening point shall be in accordance with 3.5.1 and 3.5.2 of this specification. The test shall then be repeated for a 4 G_z per sec onset/offset rate. The above test, at both onset/offset rates, shall be repeated for 70, 150 and 300 psig valve inlet pressures.

4.5.12.2.2 Test profile 2.

The valve inlet pressure shall be maintained at 30 psig while the centrifuge is incrementally accelerated from 1.2 G_z to G_z levels of 2, 4, 6, 8 and 10 at an onset rate of 4 G_z per sec. Each G_z level shall be maintained for 10 seconds. It shall then be incrementally decelerated to 8, 6, 4, 2, and 1.2 G_z at a rate of 4 G_z per sec with 10 seconds at each level. This shall be as shown on figure 9. The valve outlet pressure shall be in accordance with 3.5.2 through 3.5.5 of this specification. The above tests shall then be repeated with inlet pressures of 70, 150 and 300 psig.

4.5.12.2.3 Rotated mounting.

Mount the test valve so that the direction of valve's acceleration sensor shall be 22 degrees from the direction of the applied acceleration, in the direction of the applied acceleration, in the direction of the relief valve (see figure 10). With the inlet pressure at 70 psig, accelerate the test valve at an onset rate of 0.1 G_z per sec, to 6 G's, hold at this level for 10 sec, and decelerate at an offset rate of 0.1 G_z per sec to normal gravity. The valve outlet pressure, with respect to the acceleration in the direction of the acceleration sensor, shall be as specified in 3.5.2. The valve acceleration shall be determined by multiplying the actual acceleration by Cosine (22°).

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4.5.12.2.4 Repeat test.

Repeat the tests of 4.5.12.2.3 using an onset and offset rate of 4 G_z per sec and ensuring that the valve's performance meets the requirements of 3.5.2

4.5.12.2.5 Test for anti-chatter.

The valve shall be checked for anti-chatter by mounting an MXU-804/A or an MXU-805/A in the centrifuge and connecting it to a standard anti-g suit. A profile that shall cause this valve to chatter shall be run (approximately 3-5 G_z with an onset rate of 3 G_z per sec; the inlet to the anti-g suit may need to be restricted to induce the chatter). After chatter has been induced in this valve, replace the valve with the valve under test without disturbing the rest of the set-up. Run the same profile that caused the chatter and insure no chatter shall be present in the valve under test.

4.5.13 Endurance.

A tank shall be connected as specified in 4.4.4 during this test. A source of pressurized gas shall be connected to the inlet and the valve shall be subjected to cyclic operations for the number of cycles (see 6.4.3), under the conditions, and in the order specified in table VII. The rate of cycling shall be approximately 6 cycles per minute. The outlet pressure at the maximum force of each cycle shall be in accordance with 3.5.2 for the applied force. After completing all phases of the cyclic operation, the valve shall be subjected to and pass the inspections specified in 4.5.2 through 4.5.6; except that the minimum operating force can be between 1.25 and 2.0 G_z.

TABLE VII. Conditions of cyclic operation.

Inlet Pressure (psig)	Inlet Air Temperature (°F)	Applied Force (G_z)	Number of Cycles
150	70	1 to 6 to 1	5,000
150	70	1 to 9 to 1	15
300	400	1 to 6 to 1	1,000
300	400	1 to 9 to 1	15
150	400	1 to 6 to 1	4,000
150	70	1 to 6 to 1	5,000

4.5.14 Disassembly and examination.

At the conclusion of all inspections specified herein, the valve shall be disassembled and examined. The valve shall not show evidence of excessive deterioration or wear due to testing which might affect the valve performance. The valve shall be disassembled and examined after any test in which valve failure indicates faulty construction.

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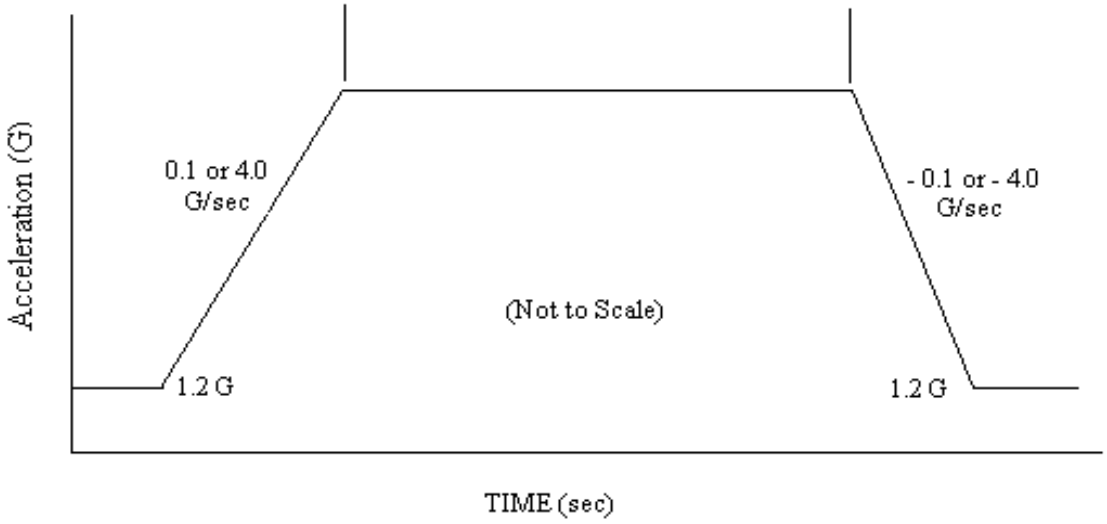


FIGURE 8. Test profile 1.

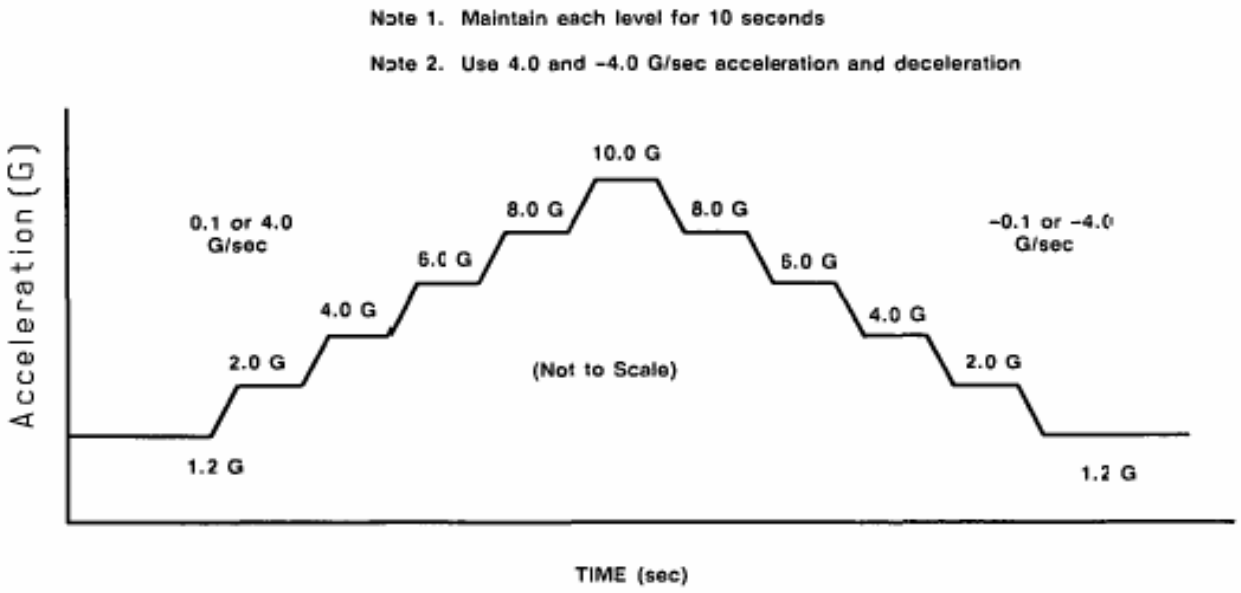


FIGURE 9. Test profile 2.

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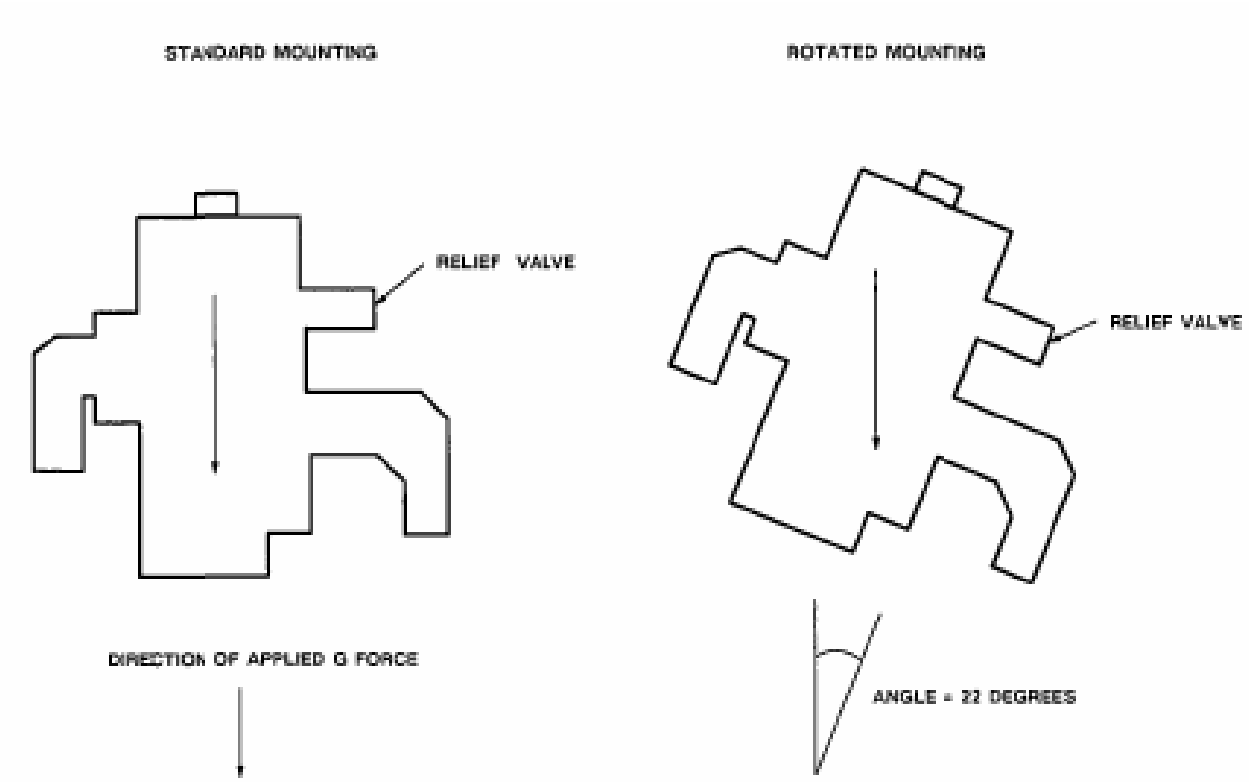


FIGURE 10. Rotated mounting for centrifuge testing.

5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

The pressure-regulating valves are intended for use in high performance aircraft. Valves designated VAU-45(V)3/A are to be used in the F-15 aircraft with or without PBG capability; valves designated VAU-45(V)2/A are to be used in the F-16 aircraft having PBG capability; and valves designated VAU-45(V)1/A are to be used in A-10 aircraft which are not going to have PBG capability. The VAU-45(V)1/A can also be used in F-16s which do not yet have PBG capability. The valves sense change in acceleration force and provide pressure for the operation of aircrew members' anti-G suits and g-compensated oxygen regulator (if applicable) in accordance with these changes.

6.2 Acquisition requirements.

Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Part number required (see 3.8).

c. When first article is required (see 3.1).

d. Centrifuge testing, if required (see 4.5.12).

e. Packaging requirements (see 5.1).

6.3 First article.

When a first article inspection is required, the item should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of four valves. Two valves shall be subjected to the verifications of 4.5.1 through 4.5.11. The other two valves shall be subjected to the verifications of 4.5.1 through 4.5.6 and 4.5.13. All valves shall be subjected to 4.5.14 as the final verification. At the discretion of the acquisition activity, the centrifuge tests of 4.5.12 may be done using four valves of identical design and construction. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions

6.4.1 Acceleration force.

An inertial force resulting from acceleration that acts upon an aircraft or person in the aircraft.

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6.4.2 Chatter.

Any rattling noise; vibration (or fluttering) in the outlet pressure, or vibration in the valve body or mounts. This phenomenon may be associated with delayed performance of the valve.

6.4.3 Cycle.

A cycle shall consist of varying the force on the valve from 1 G_z to the maximum specified for the specific cycle and back to 1 G_z.

6.4.4 G-force.

An inertial force on an object resulting from an applied acceleration due to gravity or reaction to a change of direction in unit of gravitational acceleration.

a. $\pm G_y$. Acceleration acting across a body perpendicular to its long axis in a side to side direction. Examples are a pilot in a high performance aircraft executing an uncoordinated turn and exposure to lateral buffeting.

b. $+G_z$. A positive acceleration acting along the z-axis of a body. Examples are aircraft recovery from a dive and turning maneuvers.

c. $-G_z$. A negative acceleration acting along the negative z-axis of a body axis. Examples are flying on outside loop and a nose-over.

6.4.5 Positive breathing for G (PBG).

Breathing gases are supplied under pressure to pilot's mask and at the same pressure to a vest which provides counter pressure to the chest. The pressure provided is dependent on the G_z level. A pressure sense line connects the anti-g valve to a g-compensated regulator.

6.5 Part or identifying number (PIN).

The part numbering system shall be as follows:

a. Valves designated VAU-45(V)1/A shall be identified M87255-1.

b. Valves designated VAU-45(V)2/A shall be identified M87255-2.

c. Valves designated VAU-45(V)3/A shall be identified M87255-3.

6.6 Shelf-life.

This specification covers items where shelf life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27, Shelf-Life Management Manual, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/.

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6.7 Subject term (key word) listing.

Centrifuge testing

Excitation

Resonance dwell

Sinusoidal vibration

Valve, high flow positive breathing for G

Valve, relief

Valve, pressure regulators

6.8 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 11

Preparing activity:
Air Force - 11

Project No. (1660-2007-007)

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